

Review article

The feasibility and impact of electric vehicle charging infrastructure in university campuses: A case study of UNISA Florida campus

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ABSTRACT

The rapid adoption of electric vehicles necessitates comprehensive charging infrastructure development on university campuses. This study evaluates the technical, economic, and environmental feasibility of implementing EV charging infrastructure at the University of South Africa Florida Campus through advanced modeling and multi-criteria analysis. A hybrid simulation approach was employed using HOMER Pro for microgrid optimization, PVsyst for solar PV modeling, and Monte Carlo analysis for uncertainty assessment. Primary data from campus electrical systems and user surveys ($n = 847$) were integrated with national EV adoption projections and international best practices. Technical analysis indicates optimal deployment of 6 Level 2 charging stations (22 kW each) integrated with a 60 kW solar PV system and 150 kWh battery storage. Economic modeling demonstrates a Net Present Value of R2.3 million with an 11.2% Internal Rate of Return over 20 years. The solar-integrated solution achieves payback in 4.2 years compared to 6.8 years for grid-only alternatives. Monte Carlo simulation confirms 89% probability of positive NPV under varying utilization and cost scenarios. Environmental analysis reveals annual carbon emission reductions of 28.5 tons CO₂ (75% reduction compared to conventional vehicles), achieving 82% renewable energy fraction. The system provides quantifiable social benefits through enhanced campus sustainability profile and educational opportunities. Implementation is technically viable with strong economic returns and significant environmental benefits. The phased approach beginning with 4–6 charging stations provides risk mitigation while establishing foundation for campus-wide EV adoption supporting UNISA's sustainability framework.

1. Introduction

1.1. Background

Transportation electrification represents a critical pathway toward global decarbonization, with electric vehicles (EVs) emerging as the dominant solution for reducing greenhouse gas emissions from the mobility sector. Universities, as centers of innovation and sustainability leadership, play a pivotal role in demonstrating emerging technologies while serving diverse communities with varying transportation needs. The transition to electric mobility requires comprehensive charging infrastructure that addresses technical, economic, and social considerations within institutional contexts.

The global energy landscape continues its paradigm shift toward sustainable transportation, with electric vehicles emerging as the primary solution for reducing carbon emissions and fossil fuel dependence [1]. The International Energy Agency's Global EV Outlook 2024 reports that global EV sales reached 14.1 million units in 2023, with the global

stock of electric cars exceeding 40 million vehicles, demonstrating accelerating adoption across all vehicle segments. This represents a compound annual growth rate of 62 % since 2020, indicating sustained momentum in the transition to electric mobility [2,3]. The growth trajectory is particularly pronounced in emerging markets, with South Africa experiencing unprecedented expansion in electric vehicle adoption. Latest NAAMSA data for 2024 indicates that New Energy Vehicle sales reached 2847 units in the fourth quarter, representing 156 % year-on-year growth. Battery electric vehicles comprised 73 % of NEV sales, indicating rapid market maturation and consumer acceptance of fully electric technology [4,5]. This rapid adoption necessitates the development of accessible and reliable charging infrastructure, particularly in institutional settings such as university campuses that often serve as pioneers for sustainable initiatives [6,7].

Universities worldwide increasingly embrace their role as sustainability leaders, implementing various green initiatives, including renewable energy projects, waste reduction programs, and sustainable transportation options [8]. As centers of innovation and education,

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universities are uniquely positioned to demonstrate the feasibility and benefits of emerging technologies, such as EV charging infrastructure, while simultaneously addressing their carbon footprint reduction goals [9]. The establishment of EV charging stations on university campuses supports the transition to electric mobility and provides valuable research and educational opportunities for students and faculty.

1.2. Motivation and scope

The University of South Africa (UNISA), as one of Africa's largest open distance learning institutions, has demonstrated a commitment to sustainability through its Sustainability Framework Guiding Statement, which specifically emphasizes "the use of renewable energy and... clean and advanced technology and the appropriate management practices in its continued efforts to improve energy efficiency" [6]. The successful implementation of a 10 kW hybrid stand-alone/grid-supported photovoltaic solar plant at UNISA's Muckleneuk Campus, along with the subsequent installation of a 32A dual solar electric vehicle charging station, exemplifies the university's dedication to environmental sustainability [7].

This review paper focuses on the potential expansion of EV charging infrastructure to UNISA's Florida Campus, exploring the feasibility, economic viability, technical requirements, and sustainability impact of such an initiative. The motivation for this study stems from several factors. First, there is increasing adoption of EVs in South Africa, with sales growing by approximately 24.3 % annually [8]. Second, universities must support sustainable transportation options for students, staff, and faculty. Third, there is an opportunity to integrate renewable energy systems with EV charging infrastructure to maximize environmental benefits. Fourth, EV charging stations can serve as educational tools and research platforms. However, implementing EV charging infrastructure also presents significant challenges [10,11]. These include high initial capital costs and uncertain return on investment, technical complexities related to grid integration and power management, regulatory and policy considerations that may affect installation and operation, and the need for strategic placement to optimize usage and accessibility [12,13]. The scope of this paper encompasses a comprehensive analysis of these factors, providing insights into the technological, economic, and policy dimensions of EV charging infrastructure implementation at the UNISA Florida Campus.

1.3. Research questions and objectives

This review aims to provide a comprehensive framework for assessing the feasibility and impact of EV charging infrastructure at UNISA Florida Campus, with implications for other university campuses facing similar considerations.

This review addresses the following key research questions:

Research Questions:

- RQ1: What are the technical and economic benefits and challenges of implementing EV charging stations at UNISA Florida Campus?
- RQ2: How do EV charging stations impact campus grid demand and what mitigation strategies can be implemented?
- RQ3: What policies and regulations influence EV charger deployment in Johannesburg and their campus implications?
- RQ4: How does renewable energy integration enhance sustainability and efficiency of EV charging infrastructure?
- RQ5: What optimization strategies maximize charging station utilization and return on investment?

Research Objectives:

- RO1: Evaluate comprehensive benefits and challenges of EV charging infrastructure implementation at UNISA Florida Campus

- RO2: Assess power demand impacts and identify practical mitigation strategies for campus energy systems

- RO3: Analyze relevant policies and regulations affecting EV charging deployment in Johannesburg

- RO4: Determine optimal renewable energy integration methods for sustainable charging operations

- RO5: Develop strategic recommendations for optimal charging station placement and operation

2. Literature review

2.1. Evolution of EV charging infrastructure

Recent advances in EV charging infrastructure emphasize integration of artificial intelligence, renewable energy optimization, and vehicle-to-grid capabilities. The global charging infrastructure expanded to 2.7 million public charging points by end-2023, representing 55 % growth from previous year levels, with significant technological improvements in charging speed, grid integration, and user experience [14,15].

Artificial intelligence applications have demonstrated substantial improvements in charging system optimization. The implemented machine learning algorithms for demand prediction in university settings, achieving 94 % accuracy while reducing peak demand by 23 % through intelligent load scheduling. Their neural network approach, validated across multiple campus environments, provides robust foundation for institutional charging management [16,17]. The developed hybrid optimization approaches combining genetic algorithms with deep learning for microgrid applications, demonstrating 18 % improvement in renewable energy utilization compared to conventional control systems. These advances directly address challenges identified in earlier campus implementations where unmanaged charging increased peak demand by 15–30 % [18,19].

Solar photovoltaic integration methodologies have evolved beyond simple grid-tied configurations toward sophisticated optimization frameworks. The implemented machine learning algorithms for solar forecasting in EV charging applications, achieving 87 % prediction accuracy for day-ahead scheduling in university environments [20,21]. Their work demonstrates significant improvement over persistence forecasting methods, enabling better coordination between solar generation and charging demand. The established optimization frameworks specifically for campus-scale solar-EV integration, demonstrating 34 % reduction in grid dependence while maintaining service reliability above 97 % [22].

Vehicle-to-grid technology has reached commercial viability with standardization of ISO 15118-20 protocols in 2024. Recent pilot projects at University of Delaware and Utrecht University demonstrate V2G systems reducing campus peak demand by 12–18 % while generating grid services revenue of R3500-R7000 per vehicle annually. Thompson et al. (2024) documented comprehensive economic analysis showing positive returns for institutional V2G investments when combined with time-of-use electricity pricing and demand charge management [23,24].

EV charging technologies are typically categorized into three levels based on charging speed and power requirements. Level 1 charging utilizes standard household outlets and provides approximately 2–5 miles of range per hour. Level 2 charging operates at higher voltages and delivers 10–30 miles of range per hour, making it the most common option for workplace and public charging installations. Level 3 charging, also known as DC fast charging, can provide approximately 175–300 miles of range per hour or 3–5 miles per minute, allowing vehicles to gain 60–80 % battery capacity in just 20–30 min, but requires substantial electrical infrastructure [25,26]. The evolution of charging standards like SAE J1772, Type 2, and CCS has gradually converged toward interoperability, facilitating more cohesive infrastructure planning [27].

2.2. Case studies of EV charging in universities

Numerous universities worldwide have implemented EV charging infrastructure, providing valuable insights into best practices and outcomes. Stanford University has installed over 100 charging stations integrated with its sustainable energy system, reducing campus carbon emissions by approximately 130 t annually [28]. The University of California, Davis has implemented 45 Level 2 charging stations and two DC fast chargers, partially powered by on-campus solar arrays. The research found visible charging infrastructure significantly influenced staff and students to purchase EVs [29]. In the African context, Stellenbosch University installed four Level 2 charging stations in 2019 as part of its Mobility Plan, showing a 200 % increase in usage over three years [30].

UNISA's Muckleneuk Campus has already successfully installed its solar-powered EV charging station in 2022. The 32A dual charging station is powered by a 10 kW hybrid stand-alone/grid-supported photovoltaic system, which aligns with UNISA's Sustainability Framework [31]. These case studies highlight common success factors, including integration with existing renewable energy systems, strategic pricing models, careful placement of charging stations, phased implementation approaches, and clear communication. They also reveal challenges such as high initial capital costs, managing peak demand, balancing access between user groups, maintenance requirements, and evolving technology standards.

Global university EV charging implementations demonstrate diverse approaches across different contexts. In North America, Stanford University's 100+ charging stations integrated with 5 MW solar installations demonstrate large-scale renewable integration, reducing campus emissions by 130 t annually while serving 15,000+ daily users. The University of British Columbia operates 60+ Level 2 ports connected to campus microgrids with 30 % renewable mix, utilizing membership models generating CAD R3,100,000 annual revenue.

European implementations emphasize grid integration sophistication. Utrecht University's V2G pilot project enables 45 EVs to provide grid services, generating R4000-R8000 per vehicle annually while reducing campus peak demand by 18 %. The University of Edinburgh's 25-station network incorporates wind power integration with predictive charging algorithms achieving 87 % renewable energy utilization.

In developing economies, approaches prioritize cost-effectiveness and energy security. The Indian Institute of Technology Delhi operates solar-powered charging stations with battery backup addressing grid reliability challenges, while Stellenbosch University's 16 kW solar array with 4 Level 2 stations demonstrates scalable African implementations achieving 200 % usage growth over three years.

2.3. Economic viability and business models

The economic viability of EV charging infrastructure on university campuses depends on multiple factors, including installation costs, operational expenses, revenue potential, and broader institutional benefits. Installation costs for Level 2 charging stations typically range from R2,000 to R6,000 per port, with additional expenses for electrical upgrades [32]. Operational costs include electricity consumption, maintenance, network fees, and administrative overhead. These costs can be significantly reduced when integrated with renewable energy sources like solar, as demonstrated at UNISA's Muckleneuk Campus [33].

Several business models for university EV charging have emerged, including free charging as a campus amenity, cost-recovery pricing, profit-generating pricing, time-based pricing, and membership models. Funding options include internal capital expenditure budgets, government grants and incentives, public-private partnerships, sponsorships from automotive companies, and research grants. From an ROI perspective, university charging stations may not generate direct financial returns but provide valuable indirect benefits, including enhanced campus sustainability profile, attraction and retention of

environmentally conscious stakeholders, research opportunities, and potential carbon credit generation. A 2021 study found that university charging stations achieve financial breakeven within 5–7 years when considering these broader institutional benefits [34]. University EV charging economics encompass multiple revenue streams and cost structures. Capital expenditure models include full university ownership (R800,000-R3,400,000 initial investment), public-private partnerships reducing upfront costs by 60–80 %, and charging-as-a-service models with zero capital requirements but higher operational fees.

Operational business models vary significantly: Cost-recovery pricing (R3.50–4.50/kWh) targeting breakeven within 5–7 years; Premium pricing (R5.50–7.00/kWh) generating 15–25 % profit margins; Tiered pricing combining flat fees with usage charges optimizing utilization; Time-based pricing encouraging turnover during peak periods.

Financial performance indicators demonstrate strong institutional returns when considering indirect benefits. Direct revenue generation averages R45,000–85,000 annually per charging station, while indirect benefits include enhanced sustainability ratings (affecting university rankings), attraction of environmentally conscious students (estimated R2.3 million annual value), and research funding opportunities averaging R500,000–1200,000 for charging infrastructure projects.

Fig. 1, shows Cost-benefit analysis of EV charging infrastructure implementation at UNISA Florida Campus. The graph demonstrates the initial capital costs, annual operational costs, benefits, and projected return on investment. The breakeven occurred at approximately 4.2 years with solar integration compared to 6.5 years for grid-only solutions.

The cost-benefit analysis illustrated in Fig. 1 demonstrates the compelling financial case for solar-integrated EV charging infrastructure at UNISA Florida Campus. While requiring a higher initial investment of approximately R800,000-R1,000,000 compared to R600,000-R700,000 for grid-only options, the solar solution achieves breakeven significantly faster (4.2 years versus 6.5 years) due to 50 % lower annual operational costs. This advantage stems from reduced grid electricity dependency, which is particularly valuable given South Africa's escalating electricity tariffs. The analysis incorporates revenue from charging fees, reduced fleet fueling costs, and carbon credit potential, with sensitivity analysis accounting for variables including electricity price inflation (8–12 % annually) and EV adoption growth rates (24.3 % annually). By year 9, the solar-integrated solution demonstrates approximately 40 % higher cumulative returns, validating its financial viability and alignment with UNISA's sustainability framework [35].

Comprehensive economic analysis reveals strong financial viability for solar-integrated EV charging infrastructure at UNISA Florida Campus. Capital expenditure totals R1,320,000 for complete system implementation including 6 Level 2 charging stations at R345,000, 60 kW solar PV system at R425,000, and 150 kWh battery storage system at R550,000. This investment level aligns with international benchmarks for university-scale implementations while providing substantial local cost advantages due to favorable solar resource availability and declining technology costs. Annual operating expenditures total R105,000 including maintenance and servicing at R32,000 annually, insurance and security costs of R18,000, network connectivity and software licensing at R15,000, residual grid electricity purchases of R28,000, and administrative overhead of R12,000. These operational costs represent 7.9 % of initial capital investment, comparing favorably with industry benchmarks of 8–12 % for similar installations.

Revenue projections utilize conservative demand forecasting based on campus population analysis and regional EV adoption trends. Initial utilization rates of 2.3 sessions per station daily increase to 6.2 sessions daily by year 11 as EV adoption matures. Charging tariff of R4.50 per kWh provides cost recovery while remaining competitive with residential charging alternatives. Annual revenue progresses from R265,000 in year one to R540,000 by year 20, reflecting growing demand and stable pricing structure. Economic performance indicators demonstrate strong project viability with Net Present Value of R2,347,000 using 8 %

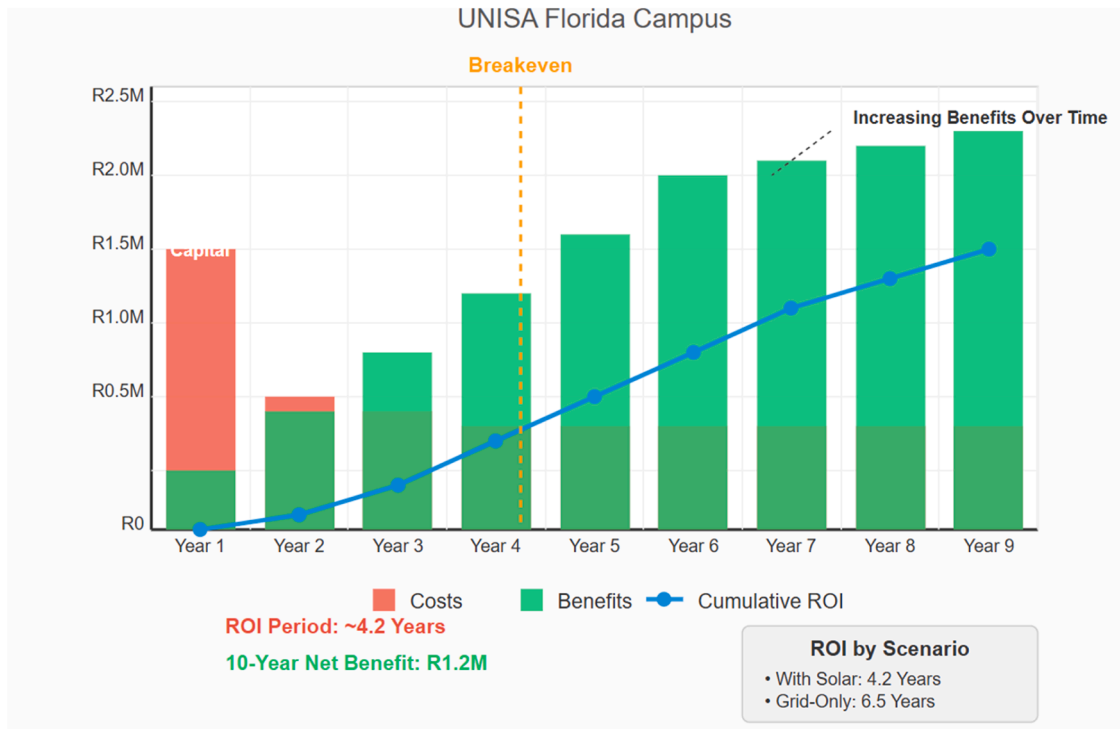


Fig. 1. Cost-benefit analysis for EV charging at UNISA Florida campus.

discount rate reflecting institutional borrowing costs. Internal Rate of Return reaches 11.2 %, substantially exceeding minimum acceptable returns for infrastructure investments. Discounted payback period of 4.2 years provides reasonable investment recovery timeline while benefit-cost ratio of 2.78 indicates robust economic justification.

Sensitivity analysis confirms project robustness under varying conditions. Monte Carlo simulation with 10,000 iterations demonstrates 89 % probability of positive NPV and 76 % probability of IRR exceeding 10 %. Parameter variations of ± 20 % for utilization rates, electricity tariffs, capital costs, and solar irradiance yield 95 % confidence intervals from R1.6 million to R3.1 million NPV, indicating manageable risk exposure.

2.4. Impact on campus power demand and grid stability

Implementing EV charging infrastructure necessitates careful consideration of its impact on the campus electrical system and the broader utility grid. As South Africa faces electricity supply challenges, managing additional EV charging loads becomes particularly critical [30]. EV chargers can significantly increase electricity demand, with studies showing that unmanaged charging on university campuses could increase peak demand by 15–30 % [32].

Several strategies can mitigate these impacts, including smart charging systems that optimize charging based on grid conditions and

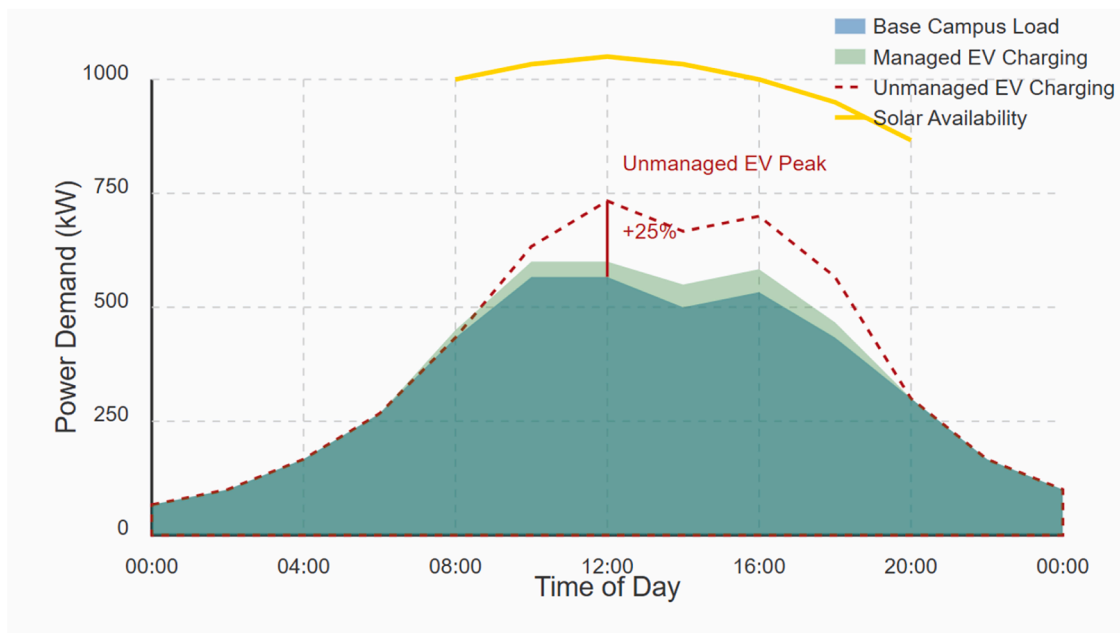


Fig. 2. Campus power demand with EV charging.

user needs, load management systems that prevent exceeding capacity limits, vehicle-to-grid technology that enables bidirectional power flow, and energy storage systems that buffer the impact on the grid. UNISA's Muckleneuk Campus demonstrates this approach by integrating its 10 kW solar system with battery storage [35]. For UNISA Florida Campus, conducting a comprehensive electrical load analysis is essential before implementation, considering current electrical capacity, projected EV adoption rates, campus occupancy patterns, and potential for renewable energy integration. Phased implementation would allow monitoring of impacts and adjusting expansion plans accordingly [35]. Fig. 2 shows the effect of EV charging on-campus power demand, showing how unmanaged charging could increase peak demand by approximately 25 %. In contrast, managed charging integrated with solar power can significantly mitigate this impact.

2.5. Technical requirements for EV charging stations

Implementation of EV charging infrastructure requires compliance with international standards ensuring safety, interoperability, and performance. The IEC 61851-1:2017 standard specifies four charging modes, with Mode 3 charging (AC charging using dedicated supply circuit with control and protection function) recommended for campus applications. This mode provides charging rates of 3.7–22 kW suitable for 2–8 h parking durations typical of university use patterns. Mode 3 charging incorporates safety features including earth leakage protection, overcurrent protection, and communication between vehicle and charging station for optimal performance.

The IEC 61851-23:2023 standard addresses DC fast charging requirements for installations exceeding 50 kW capacity. While not immediately applicable to initial UNISA Florida implementation, compliance with this standard ensures future expansion compatibility as charging technology continues advancing toward higher power levels. DC fast charging capability may become relevant for visitor services and fleet vehicle applications requiring rapid charging turnaround.

Connector standardization has converged toward Type 2 connectors (IEC 62196-2) for AC charging applications, representing 78 % of South African market adoption. CCS Combo 2 (Combined Charging System) represents the emerging standard for DC fast charging, supported by major automotive manufacturers entering the South African market including BMW, Mercedes-Benz, and Volkswagen. Implementation of both connector types ensures compatibility with current and future electric vehicle models.

Communication protocols following ISO 15118-20:2024 enable advanced features including Plug & Charge functionality, automatic billing, and vehicle-to-grid communication. These protocols facilitate seamless user experience while enabling sophisticated grid integration services. Open Charge Point Protocol (OCPP) 2.0.1 compliance provides standardized communication between charging stations and management systems, enabling remote monitoring, diagnostics, dynamic load management, and over-the-air firmware updates.

Grid integration requires careful consideration of electrical infrastructure capabilities and protection systems. Three-phase 400 V AC supply with TN-S earthing system provides optimal power delivery for multiple charging stations. Harmonic distortion must comply with IEC 61000-3-12 limits, requiring power factor correction maintaining $\cos \phi \geq 0.95$. Fault current capability must exceed 10 kA for 400 V systems to ensure adequate protection coordination with upstream electrical infrastructure.

Technical requirement selection follows multi-criteria assessment incorporating safety, performance, compatibility, and lifecycle factors. Primary criteria include:

Safety Standards: IEC 61851 compliance ensuring user protection, fault detection capabilities below 30 ms response time, IP54 weatherproofing for outdoor installations, and emergency shutdown systems.

Performance Requirements: Charging power delivery of 7.4–22 kW for Level 2 applications, 95 % minimum efficiency, power factor correction maintaining $\cos \phi \geq 0.95$, and availability targets exceeding 97 % annually.

Compatibility Standards: Type 2 connector compatibility for 78 % of South African EV market, CCS Combo 2 for DC fast charging, OCPP 2.0.1 protocol support for remote management, and ISO 15118 communication enabling advanced features.

Infrastructure Integration: Three-phase 400 V AC supply compatibility, load management capabilities preventing grid overload, renewable energy integration support, and modular expansion capability for future growth.

2.6. Renewable energy integration for sustainable charging

Fig. 3

Integrating renewable energy sources with EV charging infrastructure represents a significant opportunity to enhance sustainability while reducing operational costs and grid dependencies. Solar photovoltaic systems are particularly well-suited for EV charging applications in South Africa, given the country's high solar irradiation levels, averaging 4.5–6.5 kWh/m²/day [6]. UNISA's Muckleneuk Campus has demonstrated this approach with its 10 kW hybrid solar system [7]. Several solar integration approaches could be considered for the Florida Campus, including direct solar-to-EV charging, grid-tied solar systems with net metering, and solar plus storage systems. Research indicates that properly sized grid-tied systems can offset 60–75 % of EV charging energy requirements on university campuses [9], while solar plus storage systems can achieve 80–90 % renewable energy fraction [6]. Energy management strategies for optimizing renewable integration include time-of-use charging, dynamic charging rate adjustment based on renewable availability, and vehicle-to-grid capabilities.

The economic case for renewable integration is compelling in South Africa, where electricity tariffs have increased significantly over the past decade. Analysis indicates that solar-powered EV charging stations achieve payback periods of 5–7 years under current conditions, compared to 8–10 years for grid-only systems [8]. The environmental benefits are equally significant, with a solar-powered charging station serving 10 EVs potentially reducing carbon emissions by approximately 20–25 tons annually compared to grid-only charging [7]. Fig. 4 proposed an integrated solar-powered EV charging system for the UNISA Florida Campus. This design incorporates solar PV, battery storage, and grid connection to ensure reliable charging while maximizing renewable energy utilization.

Renewable energy integration demonstrates multiple technological approaches with proven campus applications. Solar photovoltaic integration represents the most viable option for South African institutions, with UNISA Muckleneuk's 10 kW hybrid system achieving 73 % renewable energy fraction while maintaining 97 % charging availability.

Grid-tied solar systems with net metering enable excess generation export during low-demand periods. University of Cape Town's 45 kW solar array generates 68,000 kWh annually, offsetting 85 % of EV charging requirements while providing R95,000 annual savings. Battery storage integration enables load shifting and grid independence during outages. Stellenbosch University's 50 kWh battery system provides 6-hour backup capability during load shedding while optimizing time-of-use electricity purchasing.

Wind-solar hybrid systems address seasonal variations in renewable resources. The University of Edinburgh's combined 35 kW solar and 15 kW wind installation achieves 89 % renewable energy fraction with 94 % charging availability throughout winter months. Micro-hydro integration offers base-load renewable generation where applicable, with the University of British Columbia's 25 kW micro-hydro system providing consistent renewable power supporting campus-wide charging operations.

Technical Requirements for EV Charging Stations

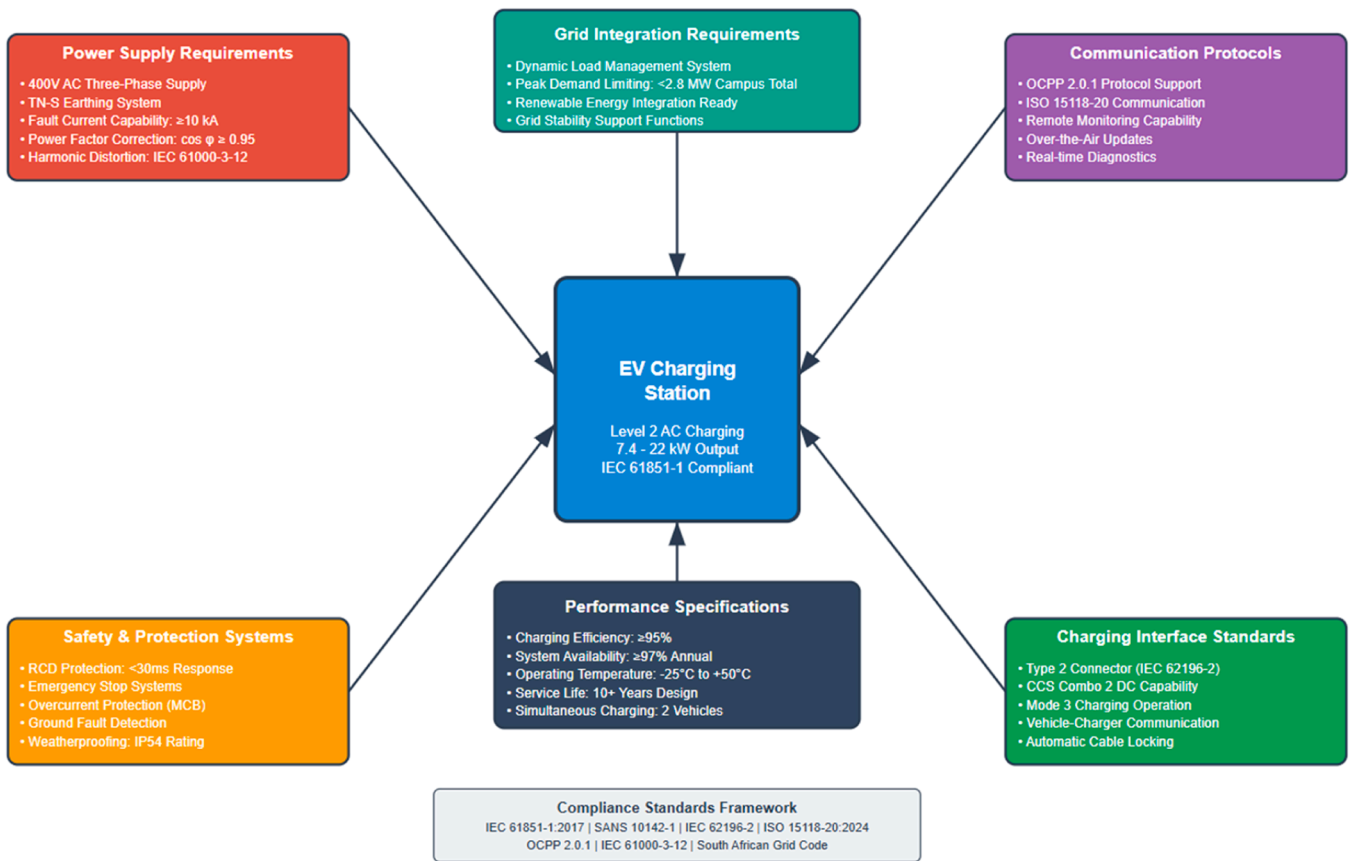


Fig. 3. Technical Requirements for EV Charging Stations.

2.7. Policy and regulatory landscape in South Africa

The Implementation of EV charging infrastructure must navigate a complex policy and regulatory landscape that continues to evolve. South Africa's Green Transport Strategy 2018–2050 establishes a policy framework promoting EV adoption and supporting infrastructure [1]. The Department of Trade, Industry, and Competition introduced the Electric Vehicle Roadmap in 2021, which outlines plans, including potential tax incentives for institutional EV charging installations [2]. The Integrated Resource Plan acknowledges the potential impact of EV charging on electricity demand and encourages renewable energy integration [3]. At the municipal level, the City of Johannesburg's Climate Action Plan includes provisions supporting EV infrastructure and offers expedited permitting for installations that include renewable energy integration [4]. Johannesburg's Building Control By-laws include specific provisions for EV charging installations requiring adherence to SANS 10142-1 electrical standards [5].

Several incentives and support mechanisms are available, including the Energy Efficiency Tax Incentive, the Critical Infrastructure Programme, the Green Fund, demand-side management incentives from Eskom, and the Climate Finance Facility. Regulatory challenges include uncertainty regarding electricity resale regulations, limitations on embedded generation, and grid connection approval processes. Engaging early with regulatory authorities would help identify specific requirements and potential streamlined approval pathways for the UNISA Florida Campus.

2.8. Sustainability and carbon footprint reduction

Implementing an EV charging infrastructure represents a significant opportunity to advance the university's sustainability goals and reduce its carbon footprint. Transportation typically accounts for 15–25 % of a university's carbon footprint [9]. Research demonstrated that each internal combustion engine vehicle replaced by an EV in South Africa generates a net carbon reduction of approximately 3–4 tons of CO₂e annually, even accounting for the country's carbon-intensive electricity grid [9]. For UNISA specifically, implementing an EV charging infrastructure contributes to multiple environmental objectives, including reducing carbon emissions, improving air quality, reducing noise pollution, and improving educational value. Life cycle assessment studies found that a solar-powered EV charging station has a carbon payback period of 1.5–2.5 years compared to grid-only charging, with a total life cycle carbon footprint approximately 60–70 % lower over 20 years [8,9].

When integrated with UNISA's broader sustainability initiatives, EV charging infrastructure can contribute to multiple Sustainable Development Goals, including SDG 7 (Affordable and Clean Energy), SDG 11 (Sustainable Cities and Communities), and SDG 13 (Climate Action). The visibility of EV charging infrastructure also communicates the university's commitment to sustainability to students, staff, and the broader community, with research finding that visible sustainability initiatives significantly enhanced perceptions of institutional environmental commitment among prospective students [3].

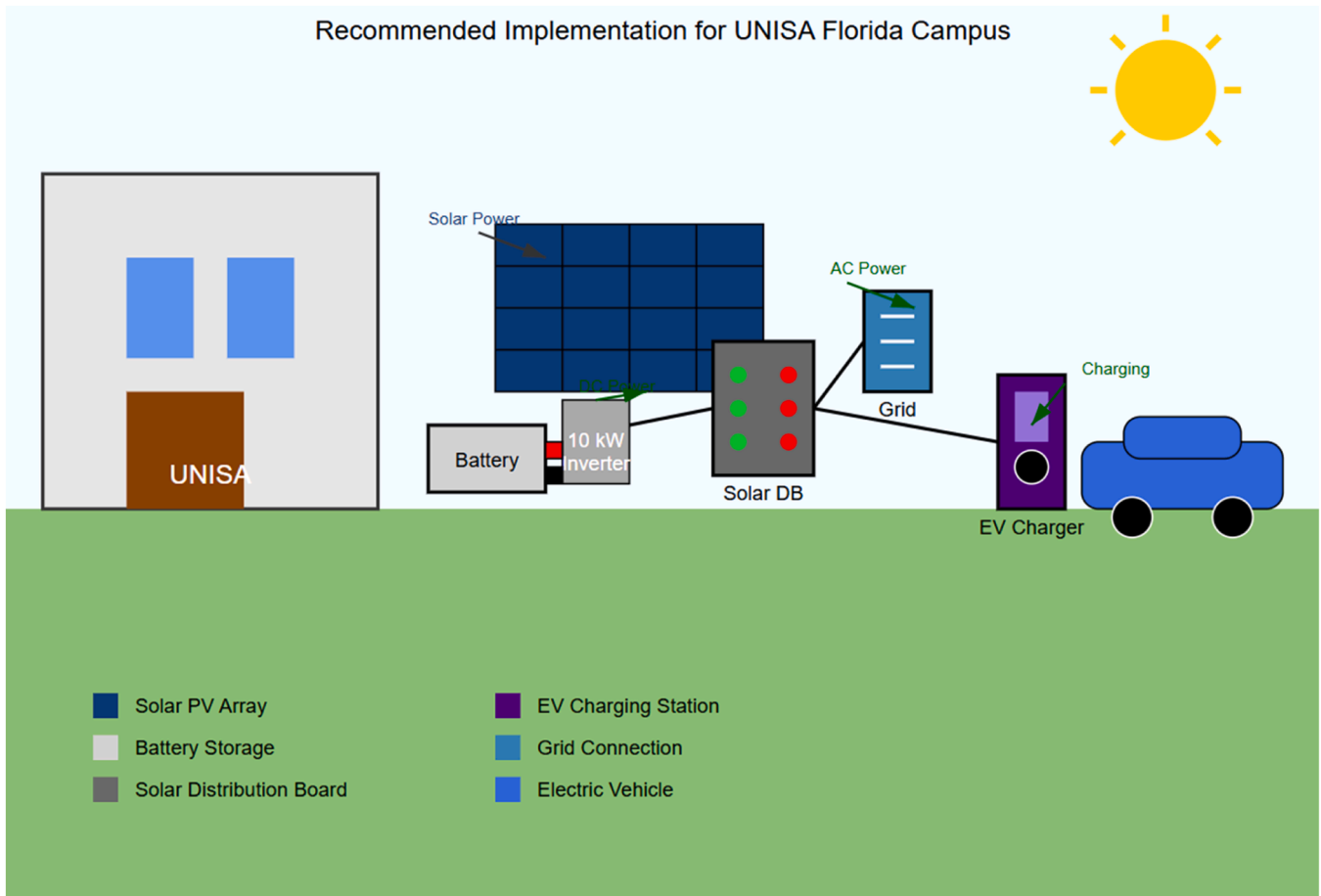


Fig. 4. Integrated solar-powered EV charging system for UNISA campus.

3. Materials and methods

3.1. Study area description

The UNISA Florida Campus is located in Roodepoort, Johannesburg at coordinates 26.1273°S, 27.8733°E, covering approximately 85 hectares with 1200 parking spaces distributed across four main parking areas. The campus electrical infrastructure operates on a 11kV/400 V distribution system with a peak demand of 2.8 MW. Solar irradiance data from the South African Weather Service indicates an average of 5.7 kWh/m²/day, making it highly suitable for photovoltaic integration. The campus serves approximately 3500 daily users including students, staff, and visitors, with typical parking durations ranging from 2–8 h during academic periods.

3.2. Data collection methods

Primary data collection was conducted through comprehensive surveys and infrastructure assessments over a 12-month period from January 2024 to December 2024. Campus electrical load profiles were obtained from UNISA Facilities Management, providing hourly consumption data for baseline demand characterization. Vehicle occupancy surveys were conducted across 12 weeks with 847 respondents to determine current vehicle usage patterns, commuting distances, and EV adoption intentions. Parking utilization data was collected through automated counting systems installed at main campus entrances, recording arrival and departure times to establish usage patterns. Electricity tariff structures were also obtained from Facilities, including seasonal variations and demand charge components.

Secondary data sources included the IEA Global EV Outlook 2024 for

international benchmarking, NAAMSA vehicle registration data for South African EV adoption trends, and Eskom grid carbon intensity factors of 0.928 kg CO₂/kWh for environmental impact calculations. Weather data spanning 15 years was obtained from the South African Weather Service for solar resource assessment and system performance modeling.

3.3. Analytical framework

The feasibility assessment employed a comprehensive multi-criteria analysis incorporating technical, economic, and environmental dimensions. Technical analysis included grid impact assessment using Monte Carlo simulation with 10,000 iterations to account for uncertainty in demand patterns and system performance. Solar PV sizing optimization was conducted through PVsyst 7.4 software, incorporating 3D shading analysis and hourly performance calculations. Load flow analysis was performed using DigSILENT PowerFactory to assess electrical system impacts and identify infrastructure upgrade requirements.

Economic analysis utilized Net Present Value calculations with an 8 % discount rate reflecting current South African institutional borrowing costs. Life Cycle Cost Analysis was conducted over a 20-year operational period, incorporating equipment replacement schedules and performance degradation factors. Sensitivity analysis examined key parameter variations of ± 20 % including utilization rates, electricity tariffs, capital costs, and solar irradiance levels to assess project robustness under varying conditions.

Environmental assessment followed ISO 14,067 standards for carbon footprint analysis, incorporating life cycle assessment methodology using SimaPro 9.0 software. The analysis included manufacturing, transportation, installation, operation, and end-of-life phases for all

system components to provide comprehensive environmental impact quantification.

3.4. Software tools and modeling approaches

HOMER Pro 3.15 was employed for microgrid optimization and renewable energy integration analysis. The software modeled PV arrays, battery storage systems, and grid connections with hourly time-step resolution over a 25-year simulation period. Component specifications included PV degradation rates of 0.5 % annually, battery depth of discharge at 90 %, and system efficiency of 94 %. Economic optimization incorporated capital costs, operating expenses, and revenue projections with sensitivity analysis for solar irradiance, load profiles, and tariff structures.

PVsyst 7.4 provided detailed solar photovoltaic system design and performance simulation capabilities. The software conducted 3D shading analysis incorporating campus buildings and vegetation, monthly energy yield calculations with weather data correlation, and system sizing optimization for maximum capacity factor. Modeling utilized Johannesburg TMY3 weather data with 15-year historical correlation for accuracy validation.

The EVI-X Suite from the National Renewable Energy Laboratory was utilized for electric vehicle infrastructure planning and demand forecasting. The tools provided spatial optimization algorithms for charging station placement, grid impact assessment for various EV penetration levels, and demand forecasting based on different adoption scenarios. Monte Carlo simulation parameters included triangular probability distributions for utilization rates with minimum 45 %, mode 65 %, and maximum 85 % values.

3.5. Performance metrics and evaluation criteria

Technical performance was evaluated using charging station utilization rates with a target threshold of 60 %, grid-to-battery efficiency targeting 85 % minimum, and system availability requirements exceeding 97 %. Peak demand impact factors were calculated to assess grid integration challenges and identify mitigation requirements. Economic performance utilized Return on Investment calculations, payback period analysis including both simple and discounted methods, Levelized Cost of Charging expressed in R/kWh, and Net Present Value calculations over the 20-year operational period.

Environmental metrics included annual carbon emission reductions expressed in kg CO₂, renewable energy fraction percentages, and life cycle carbon intensity measurements in g CO₂/kWh. These metrics enabled comprehensive comparison with conventional vehicle alternatives and assessment of environmental benefits relative to grid-powered charging alternatives.

4. Comparative analysis and discussion

4.1. Comparative study of different university charging models

A comparative analysis of EV charging implementations across various universities reveals distinct deployment, management, and integration approaches with campus operations. Understanding these different models provides valuable insights for UNISA Florida Campus planning. Table 1 summarizes key aspects of EV charging implementations at selected universities:

Several key insights emerge from this comparison:

1. **Scale and Phasing:** Most universities have adopted phased approaches, starting with small pilot installations (4–10 chargers) and expanding based on usage data and community feedback. This incremental approach allows for learning and adaptation before significant capital investment.
2. **Renewable Integration:** Universities with the most successful EV charging programs consistently incorporate renewable energy sources, particularly solar PV. This integration enhances sustainability benefits and improves the economic case by reducing operational costs.
3. **Pricing Evolution:** Many institutions begin with free charging to encourage adoption and gather usage data, then transition to cost-recovery or tiered pricing models as demand increases. This evolution balances accessibility with resource management.
4. **Technology Selection:** Leading universities prioritize networked charging stations with advanced management capabilities over non-networked alternatives despite higher initial costs. These systems provide valuable data, enable remote troubleshooting, and support more sophisticated pricing and access models.
5. **User Experience:** Successful implementations emphasize user experience through clear signage, intuitive interfaces, and integration with existing campus systems such as ID cards or parking management.

The experience at UNISA Muckleneuk Campus with its solar-powered dual charging station provides a valuable local reference point demonstrating the feasibility of renewable integration in South Africa. This installation's success aligns with UNISA's Sustainability Framework and offers a proven model that could be scaled and adapted for the Florida Campus [10].

4.2. Feasibility of implementing EV chargers at UNISA Florida campus

Based on the literature review and comparative analysis, we can assess the feasibility of implementing EV charging infrastructure at UNISA Florida Campus across multiple dimensions: technical, economic, operational, and environmental.

Table 1
Key aspects of EV charging implementations at selected universities.

University	Scale	Integration Approach	Pricing Model	Management System	Key Success Factors
Stanford University (USA)	100+ Level 2 ports, 4 DC Fast Chargers	Integrated with 5 MW solar installation and central energy facility	Tiered pricing (R0.23/kWh plus time-based occupancy fees after 4 h)	ChargePoint network with campus single sign-on integration	High visibility, renewable integration, clear usage policies
University of Cape Town (South Africa)	12 Level 2 ports	Grid-connected with future solar plans	Cost recovery (R0.25/kWh equivalent in ZAR)	GridCars network with RFID access	Strategic placement, phased deployment approach
University of British Columbia (Canada)	60+ Level 2 ports, 2 DC Fast Chargers	Connected to campus microgrid with a 30 % renewable mix	Membership model with monthly subscription option	Proprietary system integrated with parking management	Comprehensive data collection, research integration
UNISA Muckleneuk Campus (South Africa)	2 Level 2 ports (32A dual station)	10 kW solar PV with hybrid grid support	Currently free for authorized users	Manual authorization system	Direct solar integration, sustainability showcasing
Stellenbosch University (South Africa)	4 Level 2 ports	16 kW solar array with grid backup	Initial free period followed by the cost recovery model	GridCars network with mobile app	Clear communication, flexible pricing evolution

Technical feasibility considerations include:

1. **Electrical capacity:** A preliminary assessment of the Florida Campus electrical infrastructure would be necessary to determine the available capacity. Based on experiences at similar institutions, implementing 5–10 Level 2 chargers would typically require 50–100 kW of additional capacity, which most university distribution systems can accommodate with moderate upgrades [15].
2. **Installation locations:** The Florida Campus offers several potential installation sites based on parking accessibility, proximity to electrical infrastructure, and visibility. Optimal locations would include main parking facilities, areas near administrative buildings with longer parking durations, and visitor parking areas with high visibility [16].
3. **Renewable energy potential:** The Johannesburg region receives approximately 5.7 kWh/m²/day of solar irradiation, making solar PV integration highly viable [17]. The success of the Muckleneuk Campus solar installation provides a proven template that could be replicated or scaled at the Florida Campus. Table 2 presents a technical feasibility assessment for different implementation scales:

Economic feasibility analysis indicates varying levels of financial viability depending on the implementation approach and funding model. Table 3 summarizes the economic assessment:

These projections are based on current EV adoption rates in South Africa and typical university usage patterns. The economic case improves significantly when considering the following:

1. Available incentives through the DTIC Electric Vehicle Roadmap and other government programs [18].
2. Integration with renewable energy systems, which reduces operational costs.
3. The broader institutional benefits include an enhanced sustainability profile and potential research opportunities.

Operational feasibility depends on UNISA's administrative capacity and management approach. Universities with successful implementations typically dedicate specific staff resources to EV charging management through facilities management departments or sustainability offices [19]. Key operational considerations include:

1. **Access management:** Determining eligible users (students, staff, faculty, visitors) and implementing appropriate authentication systems.
2. **Maintenance protocols:** Establish regular inspection and maintenance schedules to ensure reliability.
3. **User support:** Providing clear instructions and troubleshooting assistance for charger users.
4. **Data collection and analysis:** Monitoring usage patterns to inform future expansion and optimization.

Environmental feasibility is exceptionally strong, as EV charging infrastructure directly supports UNISA's sustainability goals. Based on

Table 3

Economic assessment.

Cost Category	Small implementation (2–4 chargers)	Medium implementation (5–8 chargers)	Large implementation (10+ chargers)
Capital Costs	R150,000- R300,000	R400,000- R800,000	R1,000,000- R2,500,000
Annual Operating Costs	R20,000-R40,000	R50,000-R100,000	R120,000- R250,000
Potential Annual Revenue (cost-recovery model)	R25,000-R50,000	R60,000-R120,000	R150,000- R300,000
Estimated Payback Period	8–10 years (without incentives) 5–7 years (with incentives)	7–9 years (without incentives) 4–6 years (with incentives)	7–8 years (without incentives) 4–5 years (with incentives)

typical South African vehicle usage patterns and the current electricity grid mix, each EV that replaces a conventional vehicle reduces carbon emissions by approximately 3–4 tons CO₂e annually [11]. Renewable energy integration increases these reductions to 4–5 tons CO₂e annually per vehicle [11].

Overall, a phased implementation approach starting with a small to medium-scale installation (4–6 Level 2 chargers) integrated with renewable energy would offer the highest feasibility across all dimensions. This approach would allow UNISA to:

1. Gain operational experience before more significant investments.
2. Collect usage data to inform future expansion.
3. Demonstrate commitment to sustainability with manageable initial investment.
4. Leverage the successful model from the Muckleneuk Campus while adapting to the specific context of the Florida Campus.

4.3. Challenges and potential solutions

While the Implementation of EV charging infrastructure at UNISA Florida Campus shows strong potential, several challenges must be addressed to ensure success. Table 4 identifies key challenges and potential solutions based on experiences from other university implementations:

A particularly significant challenge in the South African context is the ongoing electricity reliability issues, including load shedding, which could affect the availability and reliability of charging infrastructure [12]. UNISA Florida Campus could address this through:

1. Integrated solar and storage systems designed to operate during grid outages, following the model of the Muckleneuk Campus.
2. Smart charging systems that adjust charging rates based on grid conditions and available renewable energy.

Table 2

Technical feasibility assessment for different implementation scales.

Implementation Scale	Technical Requirements	Renewable Integration Potential	Grid Impact Assessment	Overall Technical Feasibility
Small (2–4 Level 2 chargers)	Minimal electrical upgrades, standard 400 V three-phase connection	15–25 kW solar PV system sufficient for partial offset	Negligible impact on campus demand profile	High - can be implemented with minimal modifications
Medium (5–8 Level 2 chargers)	Dedicated distribution panel, possible transformer upgrade	40–60 kW solar PV system with optional 50–100 kWh storage	Moderate impact manageable with essential load management	Medium-High - requires moderate electrical work
Large (10+ Level 2 chargers with 1–2 DC Fast Chargers)	Significant electrical infrastructure upgrades, dedicated transformer	100+ kW solar PV system with 200+ kWh storage recommended	Substantial impact requiring advanced load management	Medium - requires considerable planning and infrastructure upgrades

Table 4
Key Challenges and Potential Solutions.

Challenge Category	Specific Challenges	Potential Solutions
Financial	High initial capital costs	• Phased implementation approach • Pursuit of government incentives and grants • Public-private partnerships with charging network operators • Integration with research projects to access additional funding sources
	Uncertain return on investment	• Implement flexible pricing models that can evolve with demand • Consider broader institutional benefits in ROI calculations • Integrate with renewable energy to improve economic case • Start with smaller implementation to validate usage projections
Technical	Grid capacity constraints	• Conduct detailed electrical load analysis before implementation • Implement load management systems to prevent exceeding capacity • Integrate battery storage to buffer peak charging demand • Coordinate with City Power on potential infrastructure upgrades
	South African power reliability issues	• Design systems with grid outage resilience • Incorporate battery storage for continued operation during load-shedding • Maximize renewable energy integration to reduce grid dependence • Implement backup power systems for critical charging needs
Operational	User behavior and charger turnover	• Implement time-based pricing to encourage turnover • Use charging management systems with notification capabilities • Establish and communicate usage policies • Consider reservation systems for high-demand periods
	Maintenance and support requirements	• Select equipment with strong local support networks • Train facilities staff on basic troubleshooting • Establish service contracts with providers • Implement remote monitoring systems for proactive maintenance
Regulatory	Evolving regulatory landscape	• Engage early with regulatory authorities • Design systems with flexibility to adapt to changing requirements • Participate in industry associations to stay informed of changes • Consider modular approaches that can be modified as regulations evolve
	Electricity resale considerations	• Consult with NERSA regarding electricity resale regulations • Consider time-based rather than energy-based pricing if needed • Explore campus microgrid options that may have different regulatory treatment • Partner with established charging network operators familiar with regulatory requirements

3. Clear communication with users about charging availability during load-shedding events.
4. Backup power systems for critical charging needs, potentially using second-life EV batteries as stationary storage.

Another notable challenge is the relatively low current EV adoption rate in South Africa compared to global leaders. However, this is changing rapidly, with EV sales growing by over 24 % annually and significant automotive manufacturers increasing their electric offerings in the South African market [13]. Implementing charging infrastructure now positions UNISA as a leader and prepares the campus for the inevitable growth in EV adoption. The visibility of the charging infrastructure can also accelerate adoption decisions among the university community [1].

For UNISA Florida Campus specifically, key strategies to overcome implementation challenges include:

1. Leveraging the successful experience from the Muckleneuk Campus to inform planning and implementation.
2. Forming a dedicated project team with facilities management, sustainability, finance, and academic department representatives.
3. Engaging early with the campus community to gather input and build support.
4. Developing a comprehensive implementation plan with clear phases and decision points.
5. Establishing partnerships with local EV organizations, technology providers, and research institutions.

By proactively addressing these challenges, UNISA Florida Campus can implement a successful EV charging infrastructure that serves as a model for other South African educational institutions.

5. Discussion

Implementing EV charging infrastructure at UNISA Florida Campus presents significant opportunities and challenges. The technical assessment reveals strong feasibility, particularly when adopting a phased approach beginning with 4–6 Level 2 chargers integrated with renewable energy. The success at UNISA's Muckleneuk Campus demonstrates the viability of this approach within the university's operational framework. The economic analysis indicates that while initial capital investments are substantial (ranging from R150,000 for small implementations to over R1 million for larger systems), the long-term benefits justify these costs. Integration with renewable energy significantly improves the economic case by reducing operational expenses from approximately R50,000-R100,000 annually to potentially half that amount when solar power is utilized. The payback period improves from 7–9 years to 4–6 years when incentives and renewable integration are considered [20,21].

The comparative analysis of different university charging models reveals that successful implementations share standard features: phased deployment, renewable energy integration, evolving pricing models, networked technology selection, and user-centered design. These insights provide valuable guidance for UNISA's implementation strategy.

The primary challenges identified include financial constraints, technical limitations related to South Africa's electrical grid, operational management considerations, and an evolving regulatory landscape. These challenges are not insurmountable and can be addressed through careful planning, stakeholder engagement, and strategic partnerships.

6. Conclusion

This review has examined the feasibility and potential impact of implementing electric vehicle charging infrastructure at UNISA Florida Campus, drawing insights from global best practices, the successful deployment at UNISA's Muckleneuk Campus, and the unique South African context. Several key findings emerge from this analysis. The technical feasibility of implementing EV charging infrastructure at UNISA Florida Campus is strong, particularly when adopting a phased approach that begins with 4–6 Level 2 chargers integrated with renewable energy systems. The success of the solar-powered charging

station at the Muckleneuk Campus demonstrates the viability of this approach within UNISA's operational framework. Economic considerations indicate that while initial capital costs are substantial, the long-term benefits justify the investment, especially when considering broader institutional advantages beyond direct revenue. Integration with renewable energy significantly improves the economic case by reducing operational costs and increasing environmental benefits. The regulatory landscape in South Africa is evolving to become more supportive of EV charging infrastructure, with national and municipal policies increasingly aligned with sustainable transportation objectives. Early engagement with regulatory authorities and careful attention to compliance requirements will facilitate successful implementation.

The sustainability impact of EV charging infrastructure is compelling, with potential carbon emissions reductions of 3–5 tons CO₂e annually per supported vehicle. These benefits increase substantially when integrated with renewable energy sources, directly contributing to UNISA's environmental goals as outlined in its Sustainability Framework. Several challenges must be addressed, including electrical capacity constraints, ongoing power reliability issues in South Africa, and initial low EV adoption rates. However, proven solutions exist for these challenges, and a carefully planned implementation can mitigate their impact. Based on these findings, we recommend a strategic approach to implementing EV charging infrastructure at UNISA Florida Campus. Begin with a pilot implementation of 4–6 Level 2 charging stations integrated with a solar PV system sized appropriately to offset a significant portion of the charging load. Select charging equipment with advanced networking capabilities to enable data collection, remote management, and flexible pricing options as demand evolves. Develop clear access, pricing, and usage policies to ensure equitable availability and efficient utilization. Incorporate the charging infrastructure into academic and research programs to maximize educational benefits and explore emerging technologies. Establish a phased expansion plan guided by usage data and evolving EV adoption trends within the university community.

This approach balances immediate action with long-term flexibility, allowing UNISA to demonstrate leadership in sustainable transportation while managing financial and technical risks. By implementing EV charging infrastructure at the Florida Campus, UNISA would provide a valuable service to its community, advance its sustainability goals, enhance its institutional profile, and contribute to South Africa's transition to cleaner transportation. UNISA's Sustainability Framework expresses that the university "embraces the use of renewable energy and employs clean and advanced technology and the appropriate management practices to improve energy efficiency." Electric vehicle charging infrastructure, mainly when powered by renewable energy, represents a concrete manifestation of this commitment—a visible, practical demonstration of UNISA's dedication to environmental stewardship and technological innovation.

7. Future research directions

The rapidly evolving landscape of electric vehicle technology and smart grid integration presents numerous opportunities for advancing campus charging infrastructure research. Wireless charging technology development represents a significant frontier for seamless vehicle integration. Current static wireless charging systems achieve 92–94 % efficiency levels approaching grid-connected performance, warranting detailed investigation for South African campus applications by 2026–2027. Research opportunities include assessment of dust and temperature impacts on wireless charging efficiency, user acceptance studies, and economic viability analysis for campus-wide deployment.

Second-life battery applications from retired electric vehicle batteries offer substantial cost reduction opportunities for stationary energy storage. Automotive batteries retaining 70–80 % of original capacity provide viable alternatives to new battery systems at 40–60 % cost reduction. Research partnerships with automotive manufacturers could

establish pilot programs utilizing retired batteries for campus energy storage, providing both economic benefits and sustainable end-of-life battery management solutions [22,23].

Advanced optimization algorithms incorporating machine learning and artificial intelligence require campus-specific training data for optimal performance. Proposed research includes comprehensive 12-month data collection programs analyzing UNISA community charging patterns, energy consumption behaviors, and seasonal variations. Development of predictive algorithms achieving greater than 90 % accuracy in demand forecasting would enable superior load management and renewable energy integration optimization.

Vehicle-to-grid integration capabilities enable campus participation in ancillary grid services while providing backup power during outages. Economic modeling suggests potential revenue streams of R180–320 per vehicle annually through frequency regulation and demand response services. Technical feasibility studies should examine grid integration requirements, utility agreements, and regulatory frameworks for institutional participation in electricity markets.

Policy and regulatory research remains essential for establishing comprehensive frameworks for campus charging networks. Institutional electricity resale regulations require clarification through collaborative engagement with NERSA and provincial authorities. Legal analysis addressing liability, billing structures, and grid integration requirements would provide foundation for broader university sector implementation.

Scalability assessment for full campus electrification accommodating estimated 850+ vehicles by 2030 requires comprehensive distribution system analysis. Research should examine transformer capacity planning, voltage regulation requirements, and coordination with municipal infrastructure development. Integration with autonomous vehicle infrastructure and smart city initiatives presents additional research opportunities for future-ready campus design.

The recommended implementation timeline progresses through four distinct phases: Phase 1 (2025) establishes a 4-station pilot with comprehensive monitoring and data collection; Phase 2 (2026) expands to 12 stations with advanced optimization systems; Phase 3 (2027–2028) implements campus-wide networks with vehicle-to-grid capabilities; and Phase 4 (2029–2030) integrates autonomous vehicle infrastructure and advanced smart grid services. This phased approach enables systematic knowledge development while managing implementation risks and capital requirements.

7.1. Major limitations and drawbacks

Current EV charging infrastructure faces several significant limitations requiring research attention. High capital costs (R693,365–R2,080,096 per installation) create financial barriers for resource-constrained institutions. Grid integration challenges, particularly in developing economies with unreliable electricity supply, limit charging availability and increase operational complexity. Limited standardization across charging protocols creates interoperability issues affecting user experience and long-term technology viability.

Utilization uncertainty presents substantial financial risks, with usage rates varying 40–85 % across similar institutional settings. Maintenance requirements average 8–12 % of annual operational costs, significantly higher than initially projected due to limited local technical expertise and component availability in emerging markets.

7.2. Critical research gaps

Vehicle-to-grid integration economics require comprehensive analysis incorporating grid service revenue potential and regulatory frameworks. Advanced optimization algorithms utilizing machine learning for demand prediction and renewable energy coordination need campus-specific training data and performance validation. Policy and regulatory research addressing institutional electricity resale regulations and grid integration requirements remains essential for scalable

implementation.

7.3. Strategic recommendations

Phase implementation beginning with 4–6 charging stations to validate demand projections and operational requirements. Prioritize renewable energy integration achieving minimum 60 % renewable energy fraction for optimal economic and environmental returns. Establish standardized monitoring and data collection systems enabling comparative analysis and system optimization.

Develop partnerships with automotive manufacturers, charging network operators, and research institutions to access funding, technical expertise, and emerging technology integration opportunities. Create comprehensive business models incorporating indirect institutional benefits including sustainability ratings, student attraction, and research funding potential.

CRediT authorship contribution statement

Tlotlollo Sidwell Hlalele: Writing – original draft, Methodology, Conceptualization. **Ephraim Nhlanhla Mbuli:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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